

Percent Change  
Formula



$$\frac{\text{New} - \text{Old}}{\text{Old}}$$



The Quadratic  
Formula



for  $ax^2 + bx + c = 0$ ,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



Arithmetic Mean  
(Average)



Sum of Terms  
Number of Terms



Median



Middle value of an  
*ordered* list



Mode



Most commonly occurring value



Arithmetic Sequence



Pattern in which you add or subtract to get the next number

Ex: 4, 10, 16, 22, 28 ....



Geometric Sequence



Pattern in which you multiply or divide to get the next number

Ex: 3, 9, 27, 81, 243 .....



$$a^b \cdot a^c = ?$$



$$a^{b+c}$$



$$\frac{a^n}{a^m}$$



$$a^{n-m}$$



$$(a^n)^m = ?$$



$$a^{nm}$$



$$a^0 = ?$$



anything raised to the zeroth power is always equal to 1



Basic Probability  
Formula



$$\frac{\text{\# of Desired Outcomes}}{\text{Total \# of Possible Outcomes}}$$



Probability of Events A and B  
Both Occurring



Prob of (A)xProb of (B)



Probability of Events  
A or B Occurring



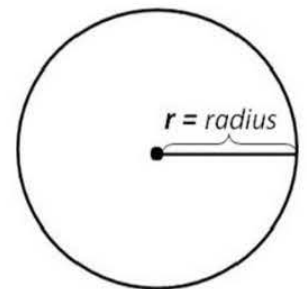
Prob of (A) + Prob of (B)



Area of a Circle  
Formula



$$\pi r^2$$

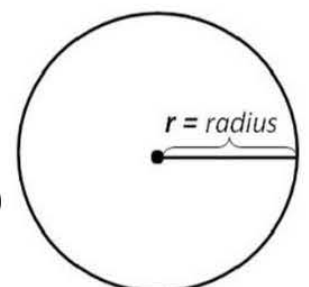


Circumference of a  
Circle Formula



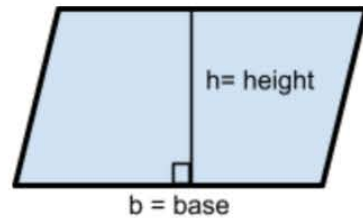
$$2\pi r$$

(or  $D\pi$ , since  $D = 2r$ )





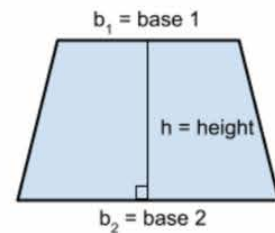
# Area of a Parallelogram



$$A = bh$$



# Area of a Trapezoid

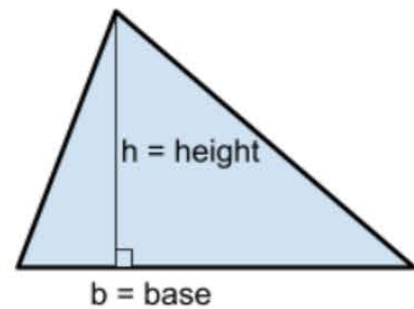


$$A = \frac{1}{2}(b_1 + b_2)h$$

(or average base times height)



# Area of a Triangle



$$A = \frac{1}{2}(b \times h)$$



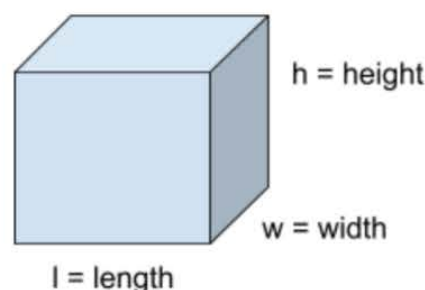
Surface Area of a 3D Shape  
(in general)



Find the area of each side (face)  
and then add them all together.



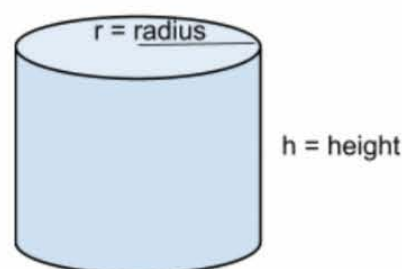
## Volume of a Rectangular Prism (Box)



$$V = l \times w \times h$$



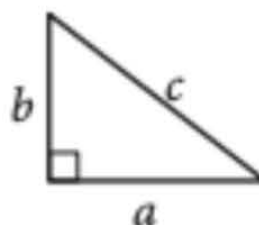
## Volume of a Cylinder



$$V = \pi r^2 h$$



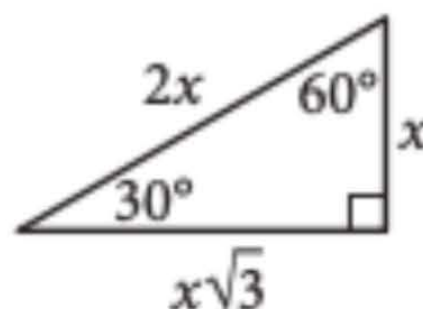
## Pythagorean Theorem



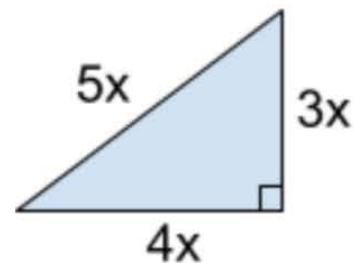
$$c^2 = a^2 + b^2$$



## Side Lengths of a 30-60-90 Triangle



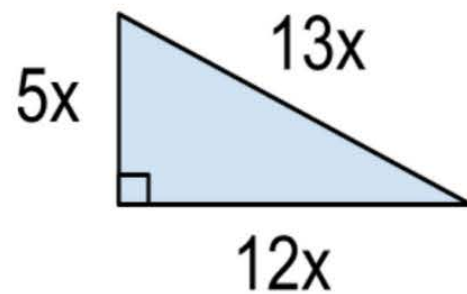
## 3 - 4 - 5 Pythagorean Triple Triangle



Examples: 3-4-5, 6-8-10,  
9-12-15



## 5 - 12 - 13 Pythagorean Triple Triangle



## SOH-CAH-TOA stands for?



S = sine  
O = opposite  
H = hypotenuse

-----  
C = cosine  
A = adjacent  
H = hypotenuse

-----  
T = tangent  
O = opposite  
A = adjacent



## Sine Equation



$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$



## Cosine Equation



$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$



## Tangent Equation



$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$



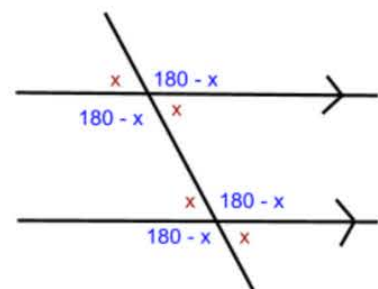
## Slope (m) Formula



$$\frac{\text{Rise}}{\text{Run}} \quad \text{or} \quad \frac{\Delta y}{\Delta x}$$



Given two parallel lines cut by a transversal, how many unique angles are created?



2 supplementary angles are created





Equation of a Line  
(Slope Intercept Form)



$$y = mx + b$$

m is slope,  
b is the y-intercept



Midpoint of  
Two Points



(average x,  
average y)



$$a^{-n} = ?$$



$$\frac{1}{a^n}$$



$$\frac{1}{a^{-n}} = ?$$



$$a^n$$



$$\sqrt[n]{a} = ?$$



$$a^{\frac{1}{n}}$$

Example:  $\sqrt[3]{7} = 7^{\frac{1}{3}}$



Changing a Log Equation  
to Its Equivalent  
Exponential Form



$$\log_2(32) = 5$$

$$2^5 = 32$$



Equation of a Circle



$$(x - h)^2 + (y - k)^2 = r^2$$

Center of the Circle:  $(h, k)$   
Radius:  $r$



## Advanced Cards Below



### Exponential Growth/ Decay Formulas



$$y = a(1 + r)^x \rightarrow \text{growth rate } r$$

$$y = a(1 - r)^x \rightarrow \text{decay rate } r$$



### Difference of Squares Formula



$$a^2 - b^2 = (a + b)(a - b)$$



### Sine & Cosine Relationship

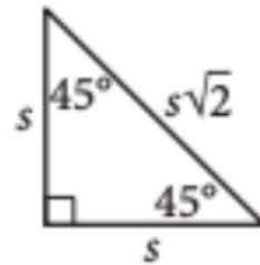


$$\sin(x) = \cos(90 - x)$$

$$\cos(x) = \sin(90 - x)$$



## Sides Lengths of a 45-45-90 Triangle



## Average Speed/ Velocity Formula



$$\frac{\text{Total Change in Distance}}{\text{Total Change in Time}}$$



## Standard form of a Quadratic (Parabola)



$$y = ax^2 + bx + c$$

Gives y-intercept **c**



## Vertex Form of a Quadratic (Parabola)



$$y = a(x - h)^2 + k$$

Vertex at point **(h, k)**





## Factored Form of a Quadratic (Parabola)



$$y = a(x - r)(x - s)$$

Gives x-intercepts **r, s**



Discriminant of the Quadratic Formula  
Value to Number of Solutions



$$b^2 - 4ac > 0, \text{ 2 Real Solutions}$$
$$b^2 - 4ac < 0, \text{ 2 Complex Solutions}$$
$$b^2 - 4ac = 0, \text{ 1 Real, Repeated Solution}$$



## The Law of Sines



$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$$

\*Used for unruly triangles when you are dealing with 2 side-angle pairs



## The Law of Cosines



$$c^2 = a^2 + b^2 - 2ab\cos(C)$$

\* Used for unruly triangles when you are dealing with 3 sides and one angle



